

Task 15 – NLP Basics: Text Classification

Objective: Build a sentiment analysis model using NLP techniques.

Dataset: IMDB Movie Reviews dataset containing positive and negative reviews.

Step 1: Text Preprocessing

- Removed HTML tags using regex.
- Removed special characters and numbers.
- Converted text to lowercase.
- Tokenized sentences into words.
- Removed stopwords using NLTK.
- Applied stemming using PorterStemmer.

Step 2: Feature Extraction (TF-IDF)

TF-IDF converts text into numeric vectors. It assigns higher importance to rare words and lower importance to common words.

Step 3: Train Test Split

Dataset split into 80% training and 20% testing.

Step 4: Model Training

Used Multinomial Naive Bayes classifier, suitable for text classification problems.

Step 5: Model Evaluation

Evaluated model using Accuracy, Confusion Matrix and Classification Report.

Achieved accuracy around 85–90%.

Interview Questions

What is TF-IDF? A technique that measures importance of words in documents.

Stemming vs Lemmatization? Stemming cuts words; Lemmatization uses dictionary meaning.